DTU Health Tech
Department of Health Technology

Multi-omics Network Analytics

Individual microbial association networks in inflammatory bowel diseases

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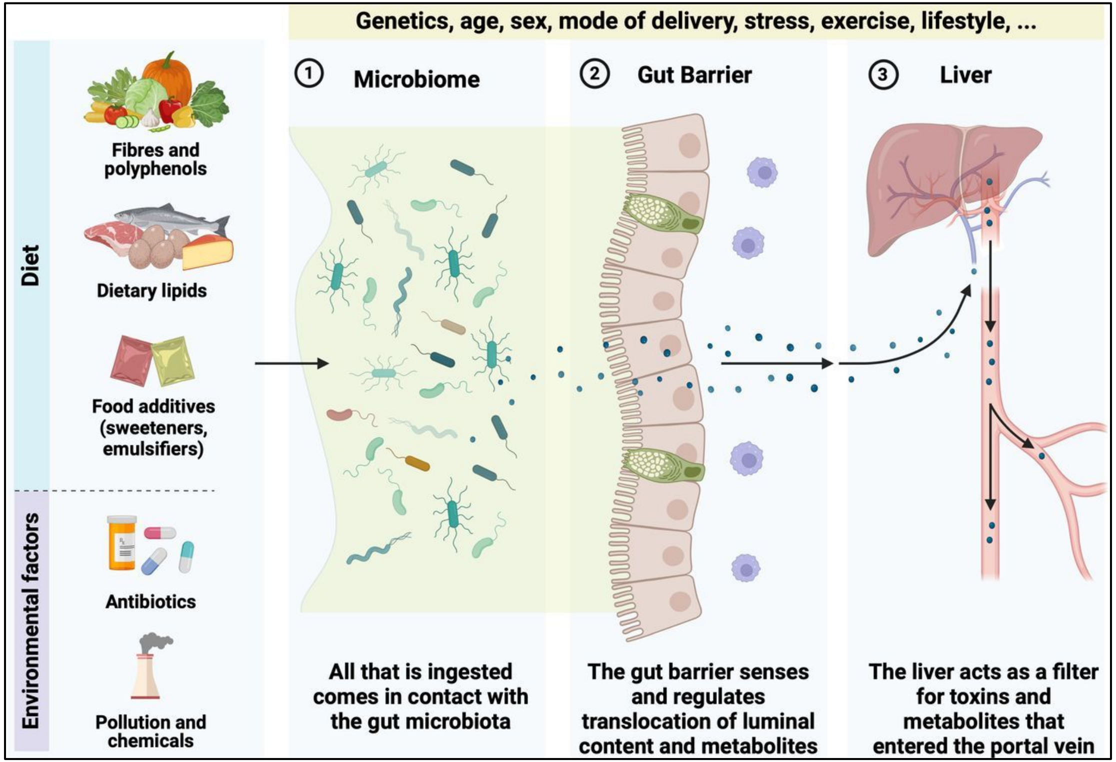
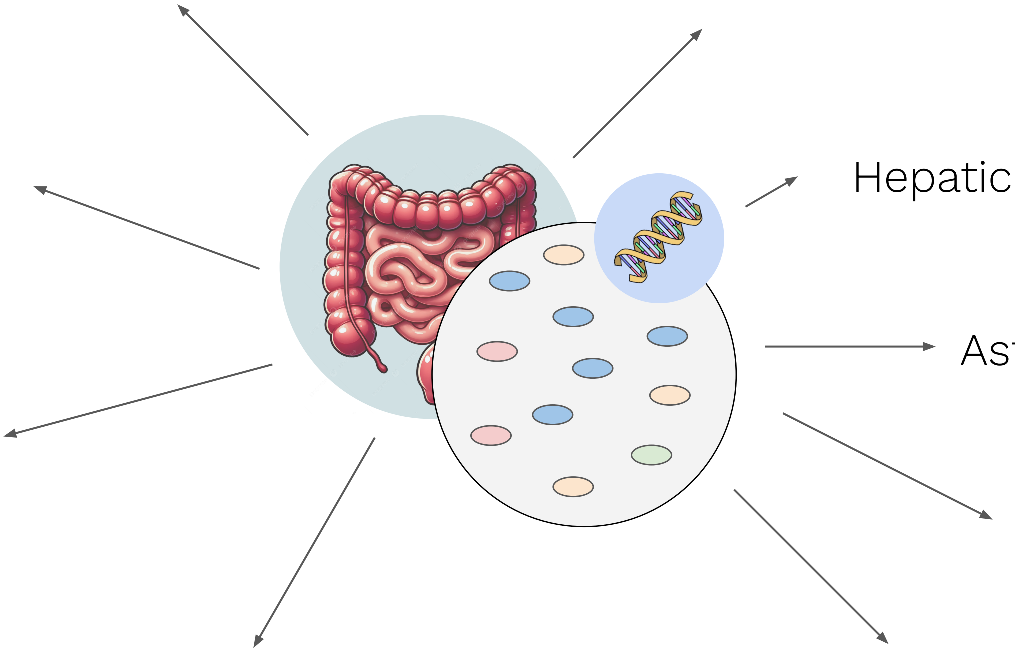
Multi-omics Network Analytics (MoNA) Group

The human gut microbiome

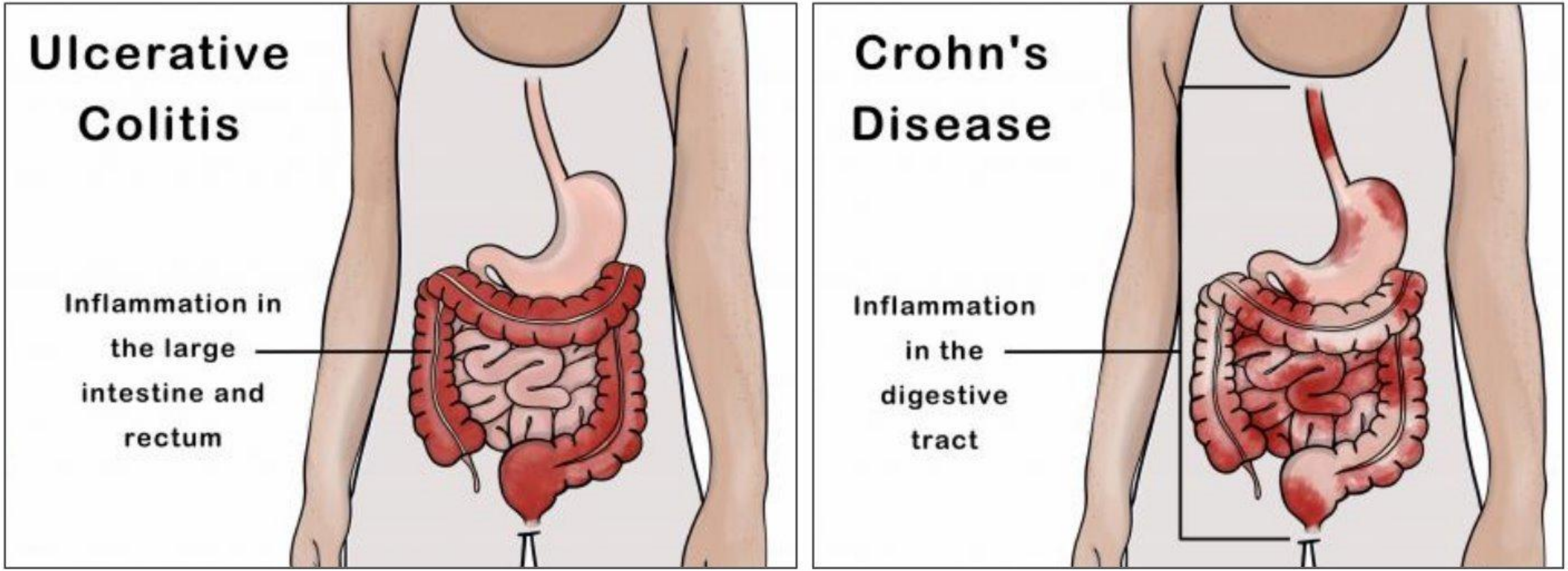


Van Hul *et al.* (2024)

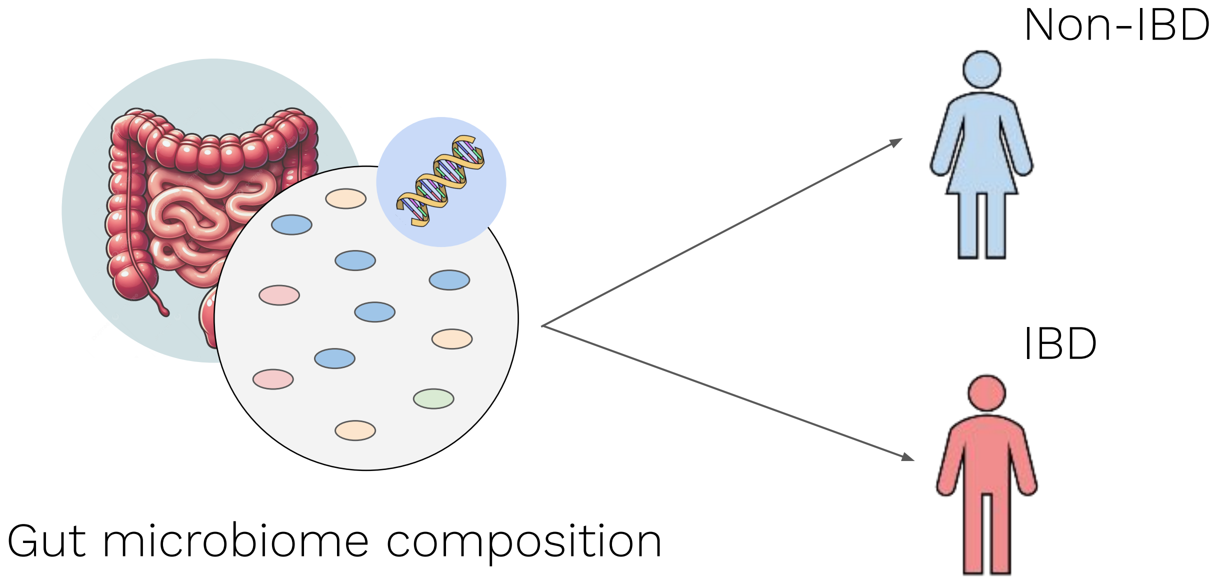
The human gut microbiome



Inflammatory bowel diseases (IBD)



Leveraging the microbiome to classify patients



Leveraging the microbiome to classify patients

Taxonomy table

	Sp1	Sp2	Sp3	Sp4
S1				
S2				
S3				

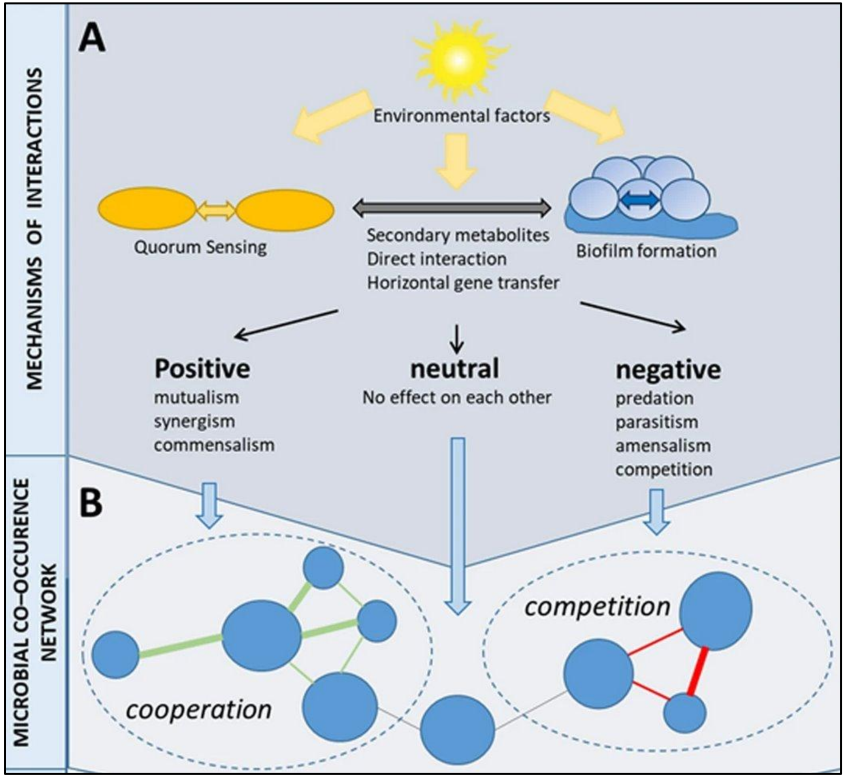
Model

f(Data)

Classification

Feature extraction

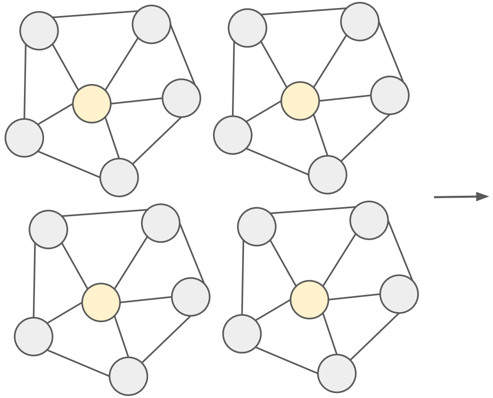
Leveraging the microbiome to classify patients



Berg *et al.* (2020)

Leveraging the microbiome to classify patients

Individual microbial association networks



Model

f(Networks)

Classification

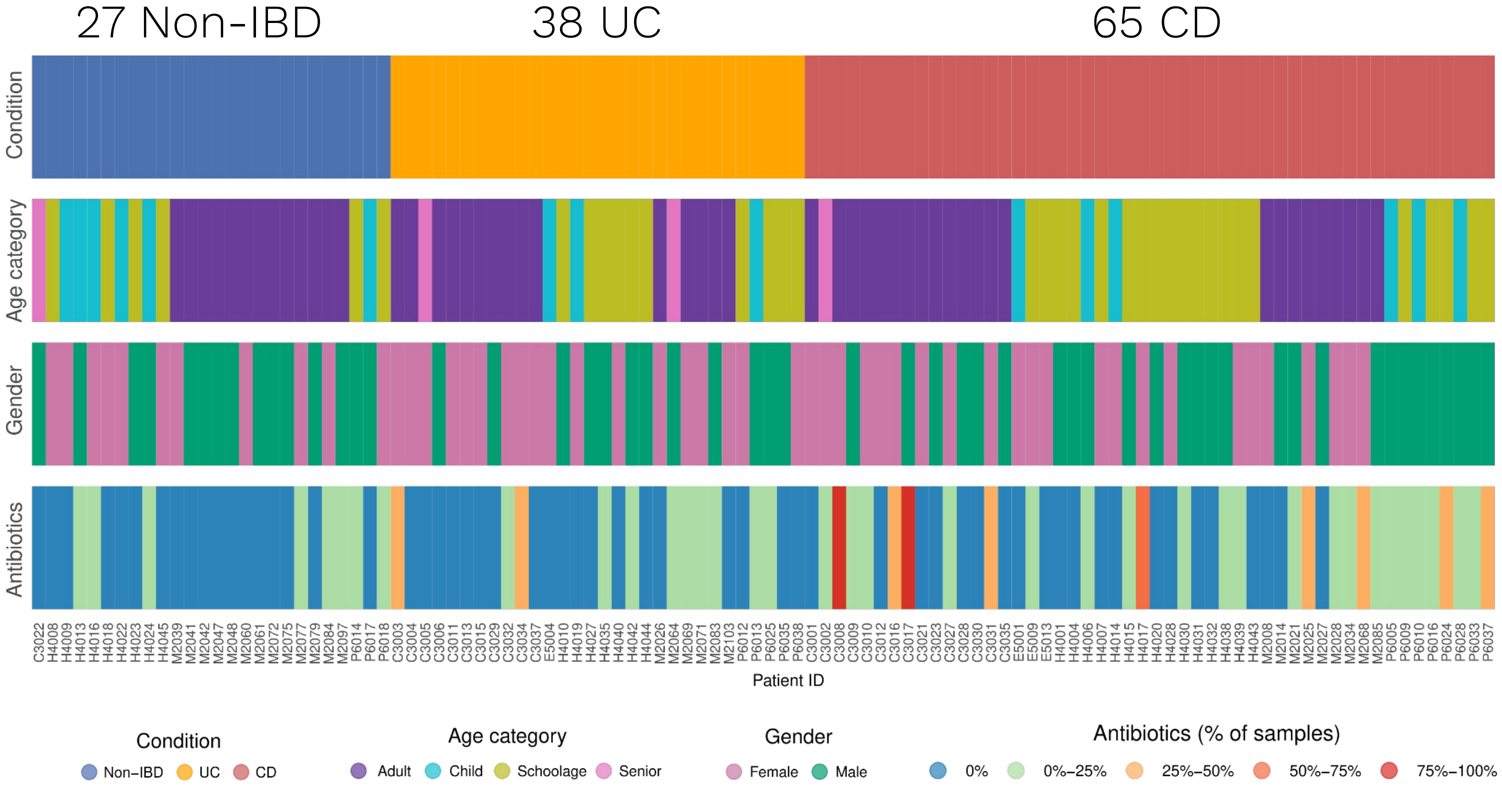
Extraction of

network-level features

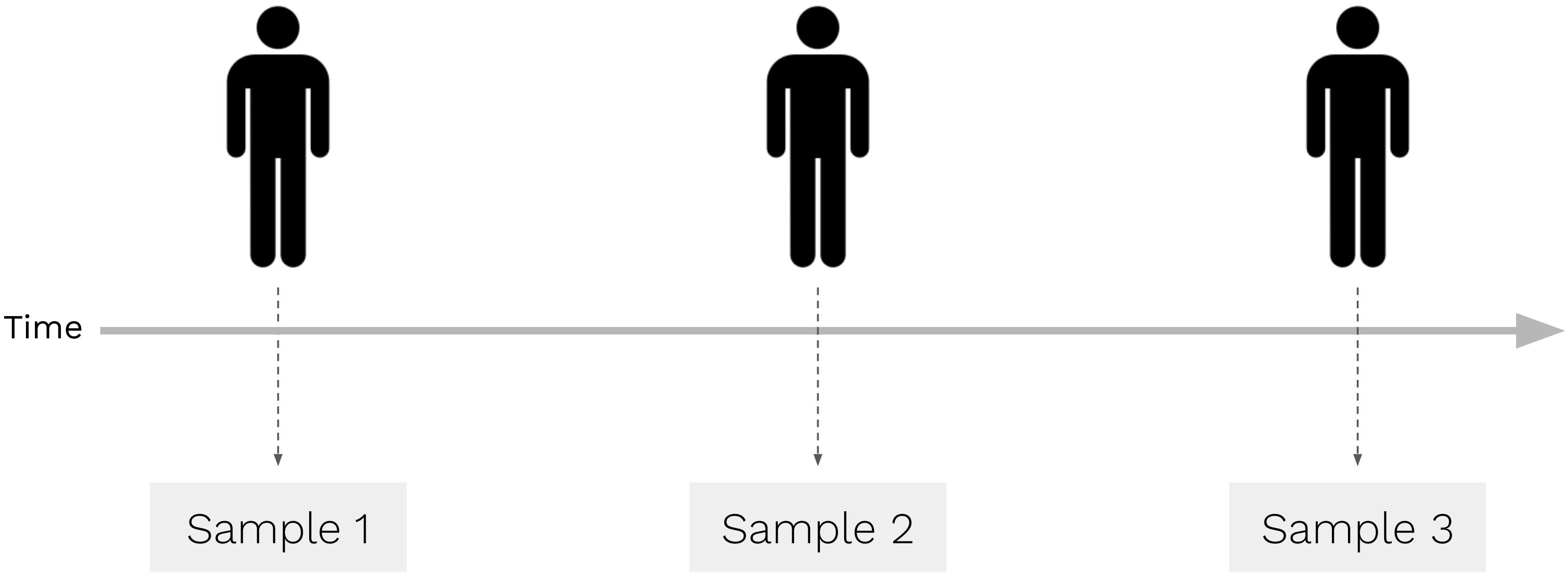
How to obtain individual microbial association networks?

Dataset source: Lloyd-Price *et al.* (2019)

HMP2/iHMP2 IBD dataset



HMP2/iHMP2 IBD dataset



Methodology overview

Species abundance table

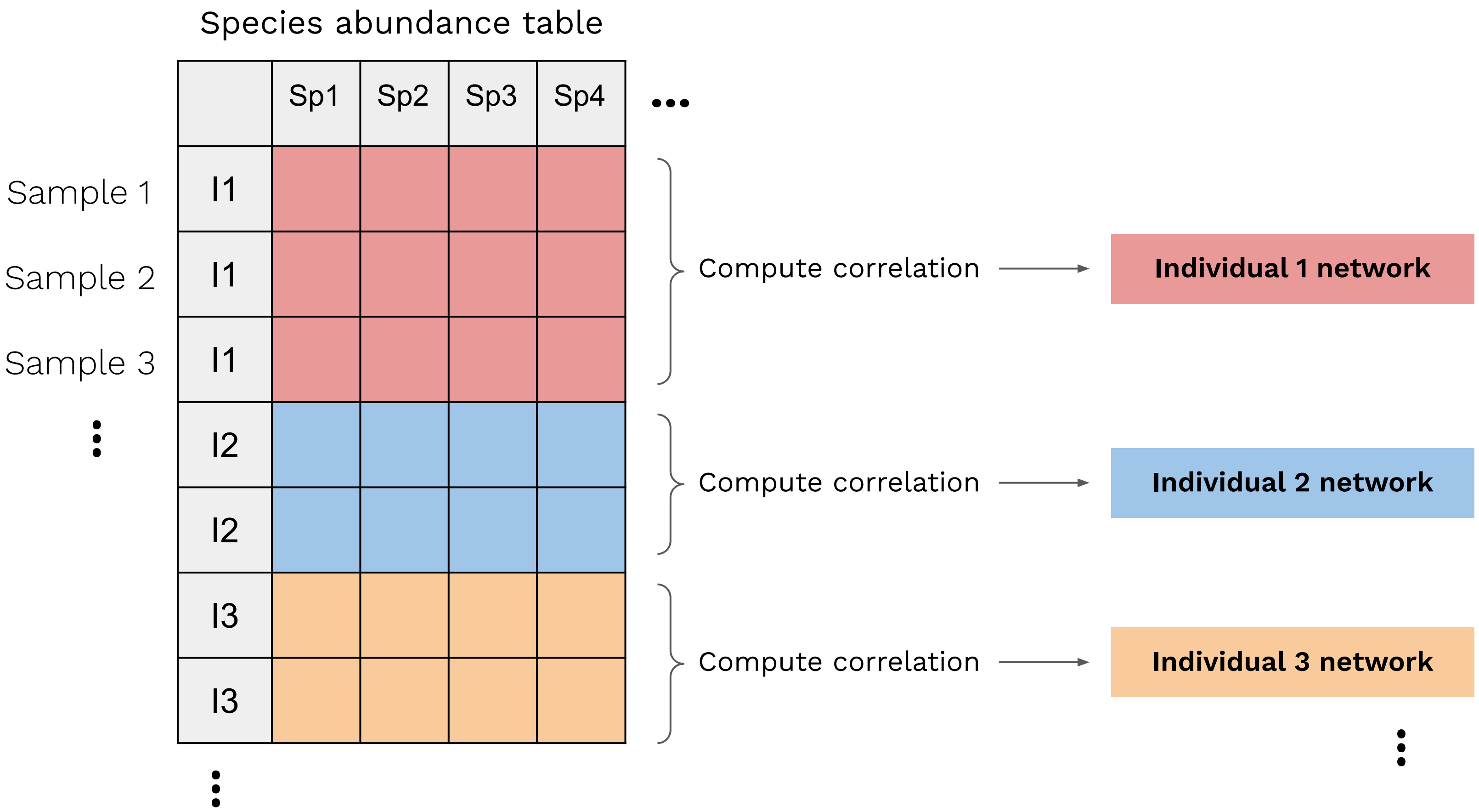
	Sp1	Sp2	Sp3	Sp4	...
I1					
I1					
I1					
I2					
I2					
I3					
I3					
⋮					

Methodology overview

Species abundance table

		Sp1	Sp2	Sp3	Sp4	…
Sample 1	I1					
Sample 2	I1					
Sample 3	I1					
⋮	I2					
	I2					
	I3					
	I3					
	⋮					

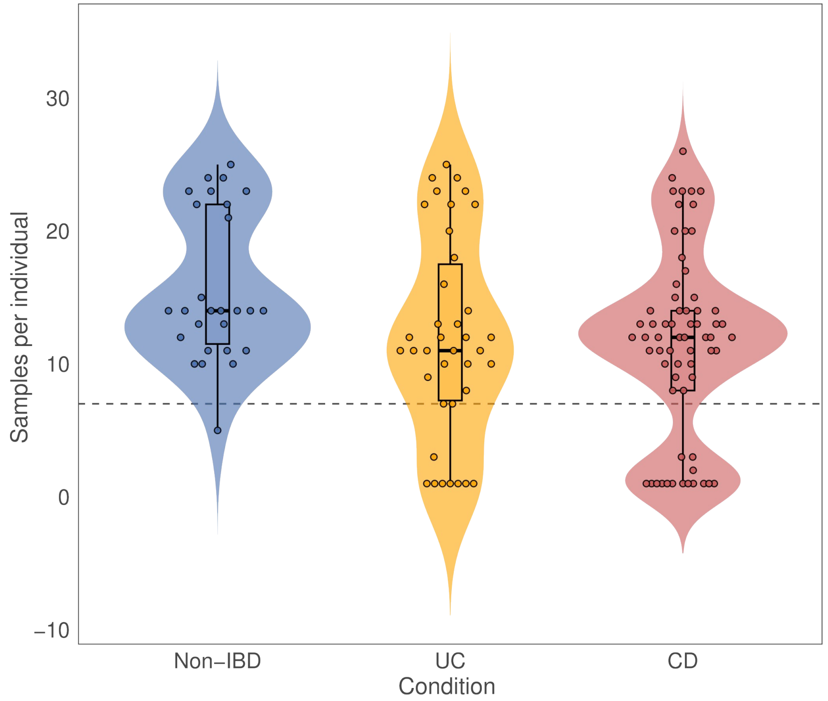
Methodology overview



Issues with the proposed network construction methodology

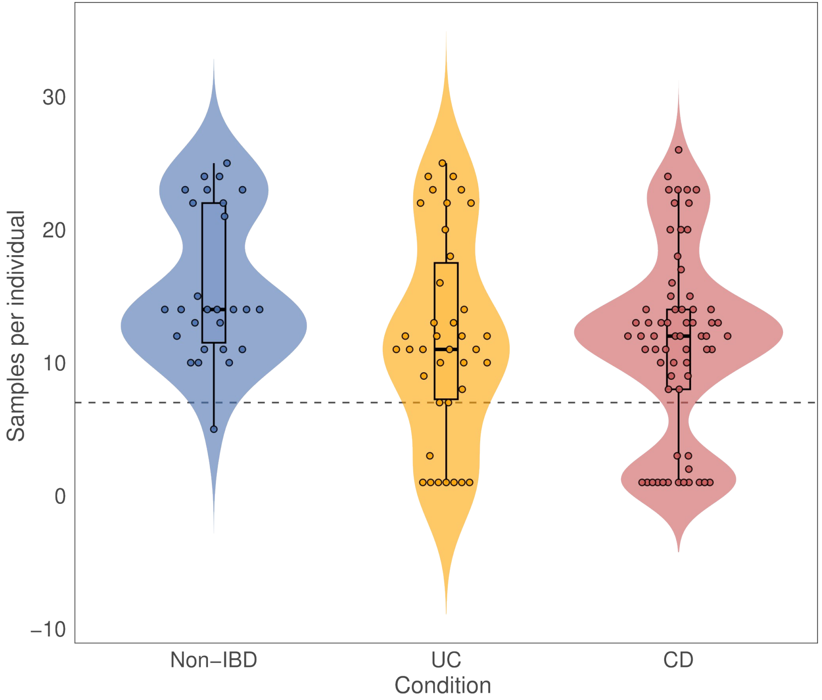
Issues with the proposed network construction methodology

Uneven number of samples across individuals



Issues with the proposed network construction methodology

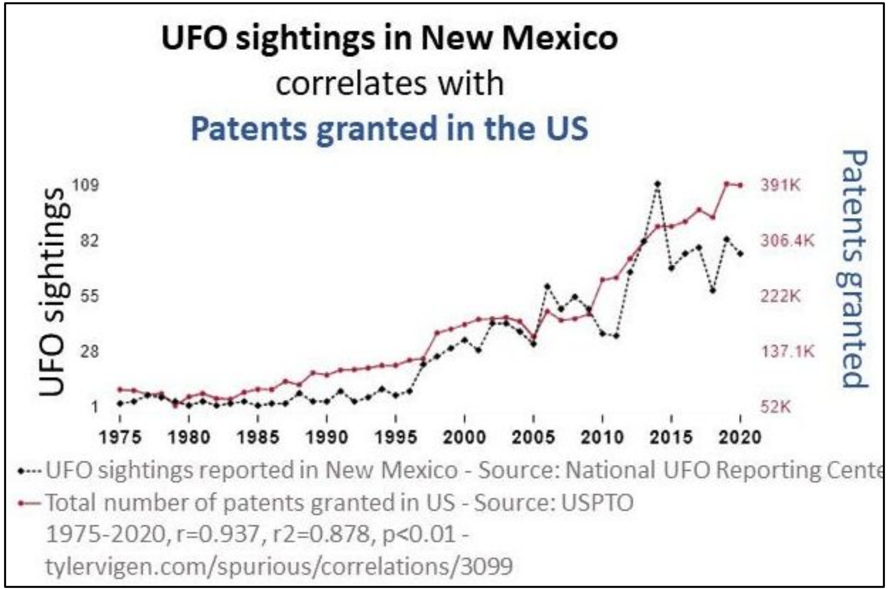
Uneven number of samples across individuals



Individuals with < 7 samples were removed from the dataset

Issues with the proposed network construction methodology

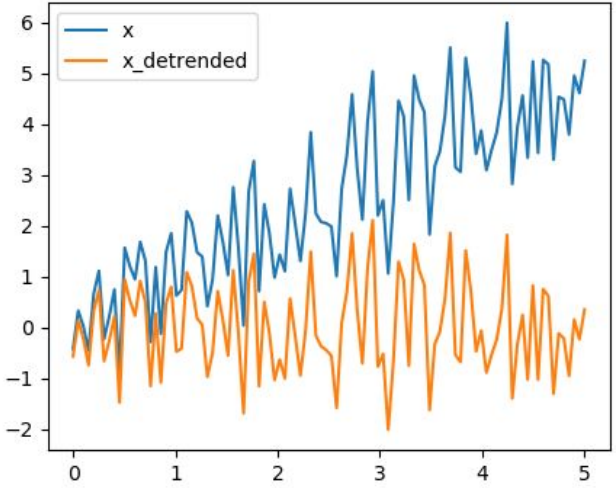
Shared temporal trends



https://medium.com/data-science/spurious-correlations-the-comedy-and-drama-of-statistics-b63bf99169d8

Issues with the proposed network construction methodology

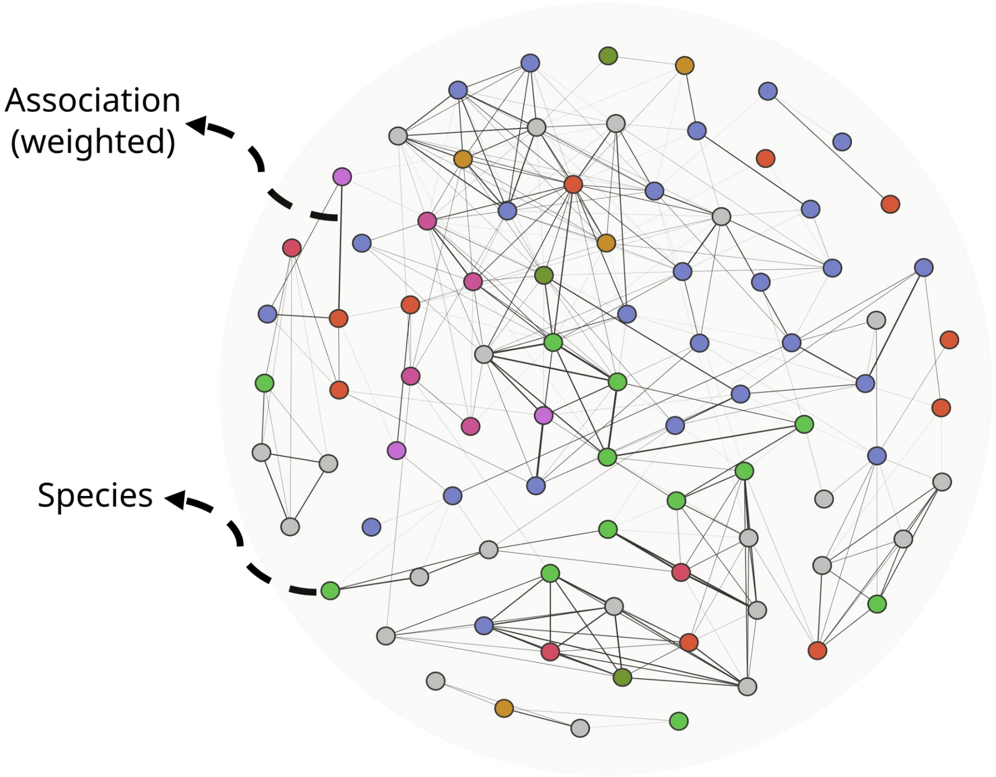
Shared temporal trends



Linear detrending was applied to each species

https://scipy-lectures.org/intro/scipy/auto_examples/plot_detrend.html

Obtaining the networks



Obtaining the networks

Taxonomic attributes

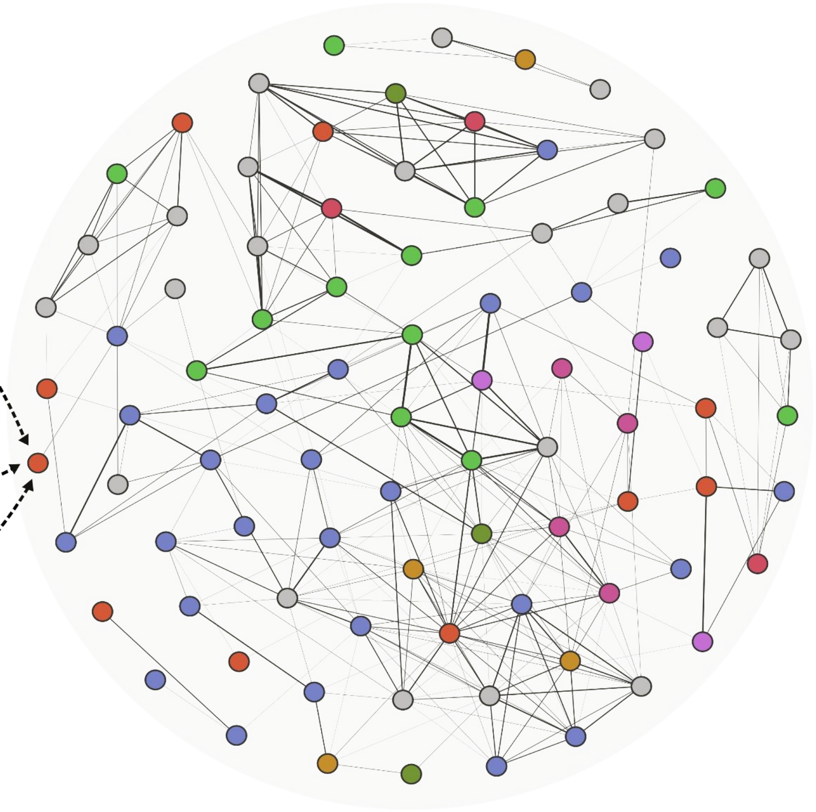
Genus 1	Genus 2	Family 1	Family 2	...
1	0	1	0	

Functional attributes

Fun. 1	Fun. 2	Fun. 3	Fun. 4	...
0	0,11	0,82	0	

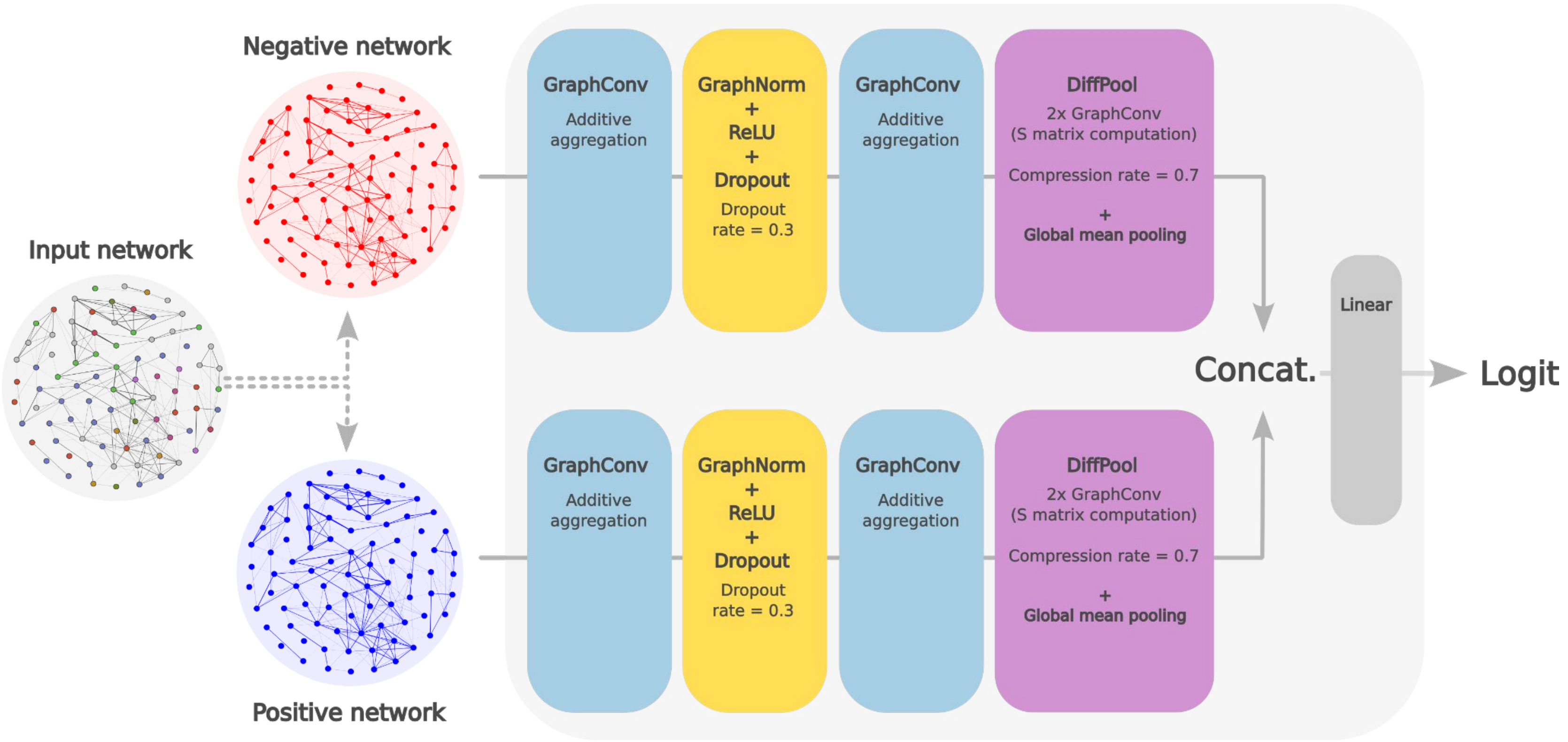
Abundance attributes

Mean CLR	Log(Var)
5,45	2,6



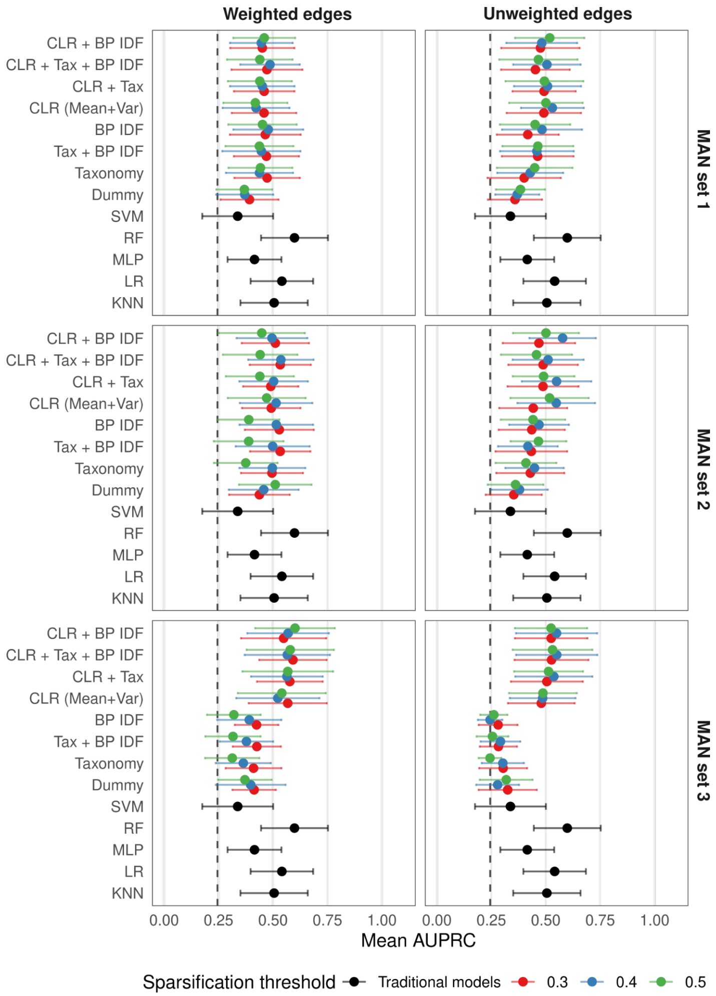
How to train a model using these networks?

Graph-based model structure



Combinations of node attributes

Traditional baseline models



Comparing performances (models)

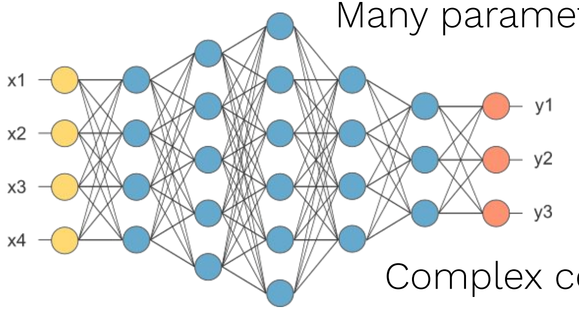
Graph-based models ↓

Random Forests ↑

Comparing performances (models)

Graph-based models

Many parameters



Complex convergence

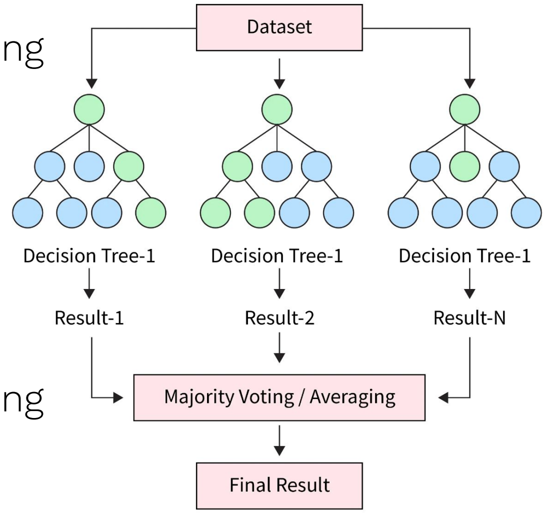
Sensitive to hyperparameters

Random Forests

Random sampling

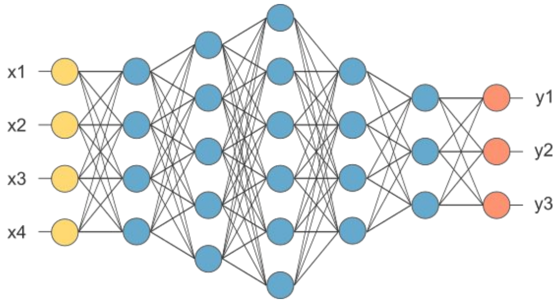
Simple algorithm

Bagging

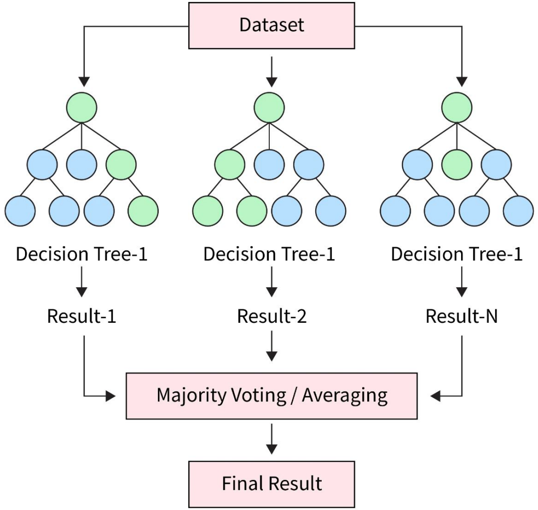


Comparing performances (models)

Graph-based models



Random Forests



106 individuals
Ratio ~ 1:3 (Non-IBD:IBD)

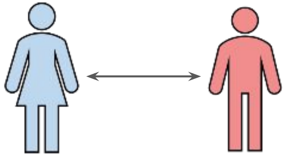
Comparing performances (node attributes)

Without node attr.

Taxonomy/Function

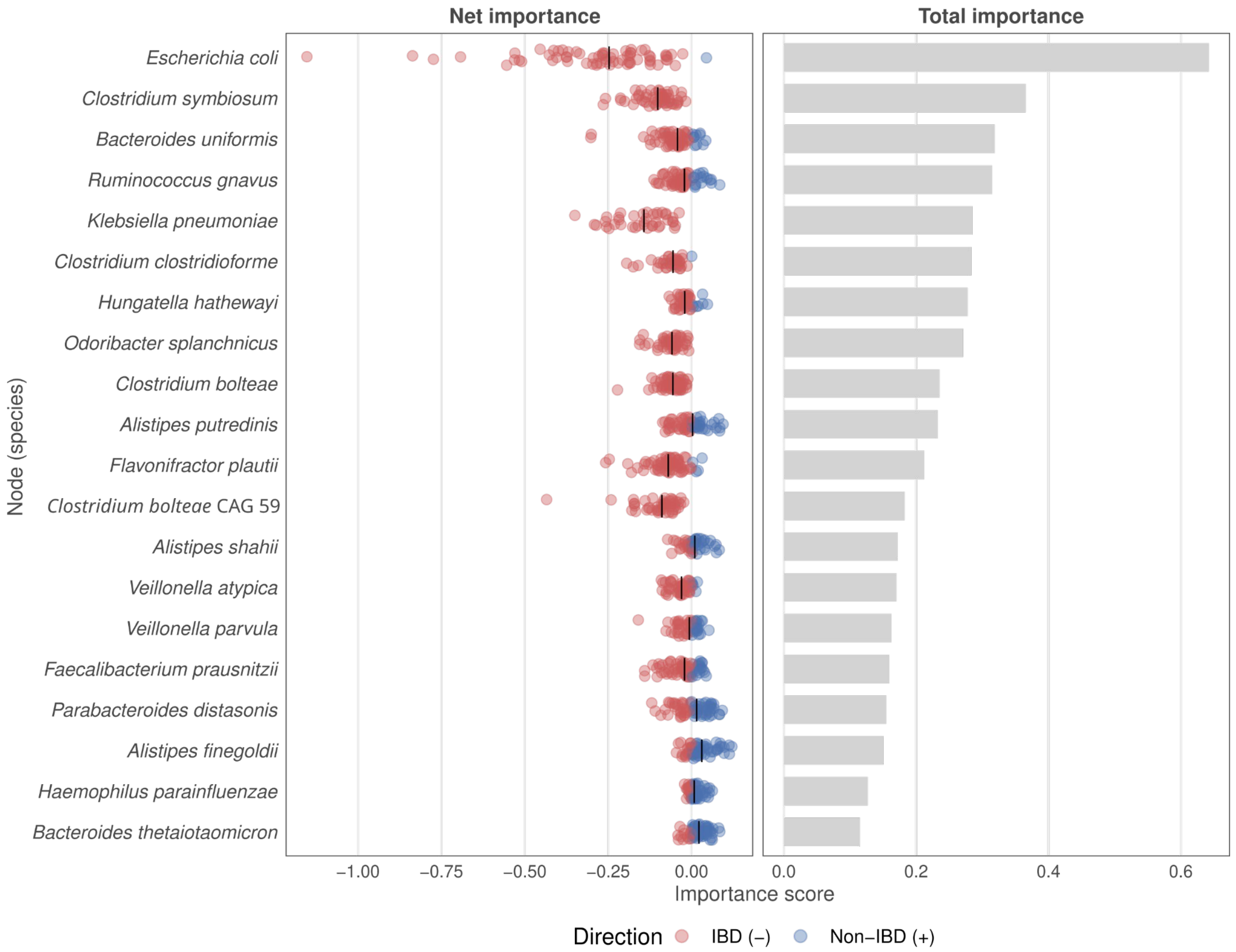
Species abundance

Variation across species

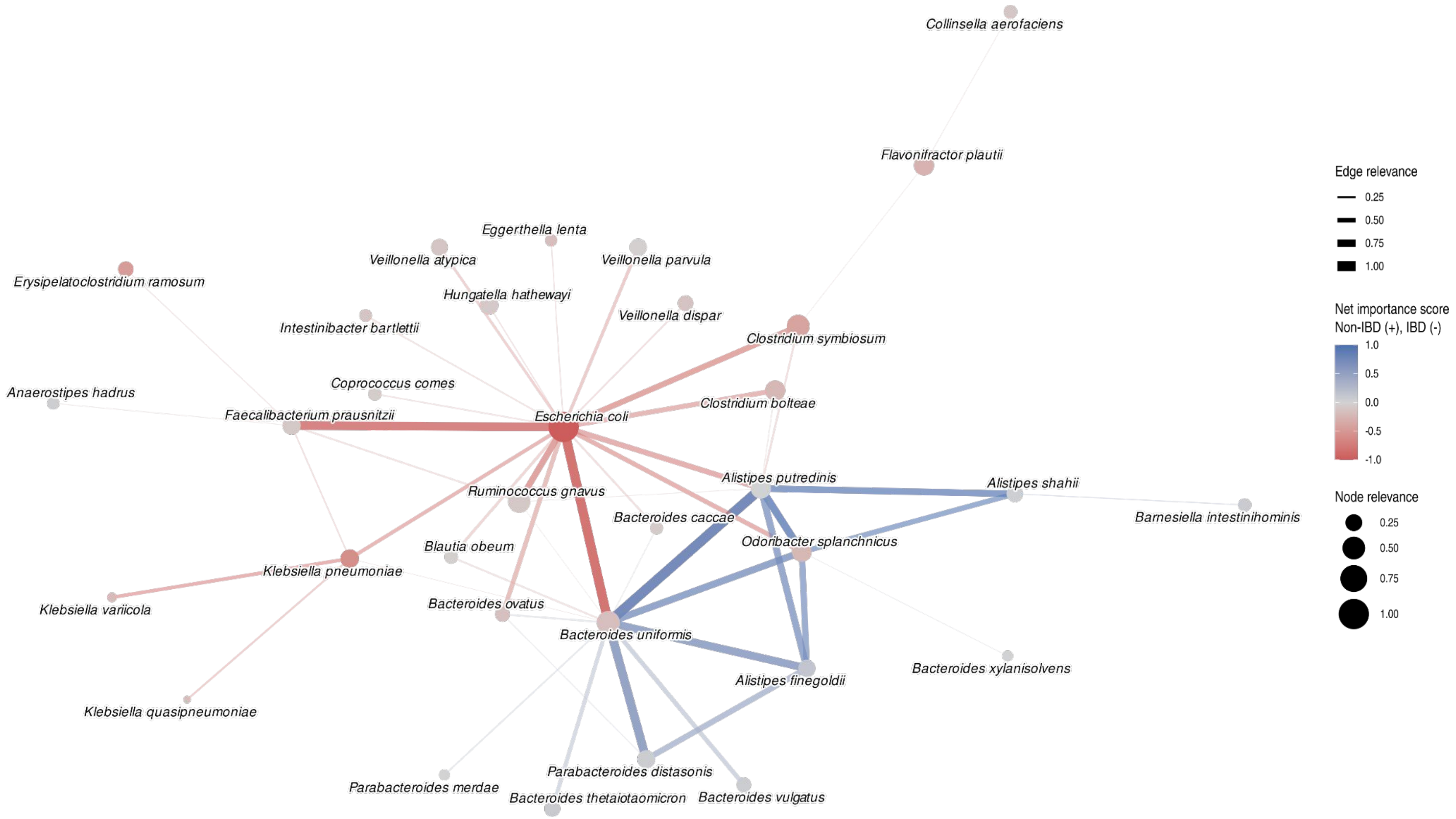


Variation across individuals

Interpreting graph-based models with Integrated Gradients

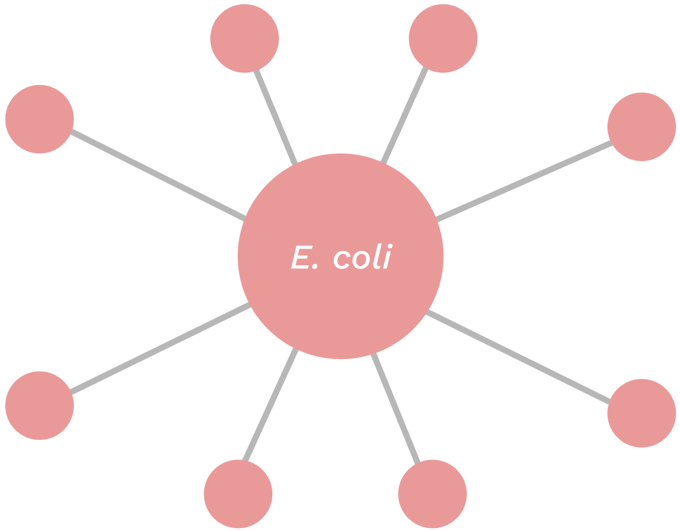


Net importance
Total importance
Escherichia coli
Clostridium symbiosum
Bacteroides uniformis
Ruminococcus gnavus
Klebsiella pneumoniae
Clostridium clostridioforme
Hungatella hathewayi
Odoribacter splanchnicus
Clostridium bolteae
Alistipes putredinis
Flavonifractor plautii
Clostridium bolteae CAG 59
Alistipes shahii
Veillonella atypica
Veillonella parvula
Faecalibacterium prausnitzii
Parabacteroides distasonis
Alistipes finegoldii
Haemophilus parainfluenzae
Bacteroides thetaiotaomicron
Node (species)
−1.00
−0.75
−0.50
−0.25
0.00
0.0
0.2
0.4
0.6
Importance score
Direction
IBD (−)
Non−IBD (+)

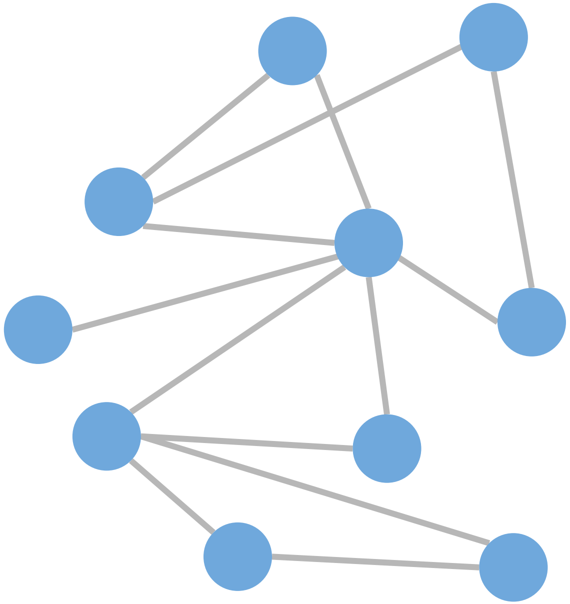


Collinsella aerofaciens
Flavonifractor plautii
Eggerthella lenta
Veillonella atypica
Veillonella parvula
Erysipelatoclostridium ramosum
Hungatella hathewayi
Veillonella dispar
Intestinibacter bartlettii
Clostridium symbiosum
Coprococcus comes
Anaerostipes hadrus
Clostridium bolteae
Faecalibacterium prausnitzii
Escherichia coli
Alistipes putredinis
Alistipes shahii
Ruminococcus gnavus
Bacteroides caccae
Barnesiella intestinihominis
Odoribacter splanchnicus
Blautia obeum
Klebsiella pneumoniae
Klebsiella variicola
Bacteroides ovatus
Bacteroides uniformis
Bacteroides xylanisolvens
Alistipes finegoldii
Klebsiella quasipneumoniae
Parabacteroides distasonis
Parabacteroides merdae
Bacteroides thetaiotaomicron
Bacteroides vulgatus
Edge relevance
0.25
0.50
0.75
1.00
Net importance score
Non-IBD (+), IBD (-)
1.0
0.5
0.0
-0.5
-1.0
Node relevance
0.25
0.50
0.75
1.00

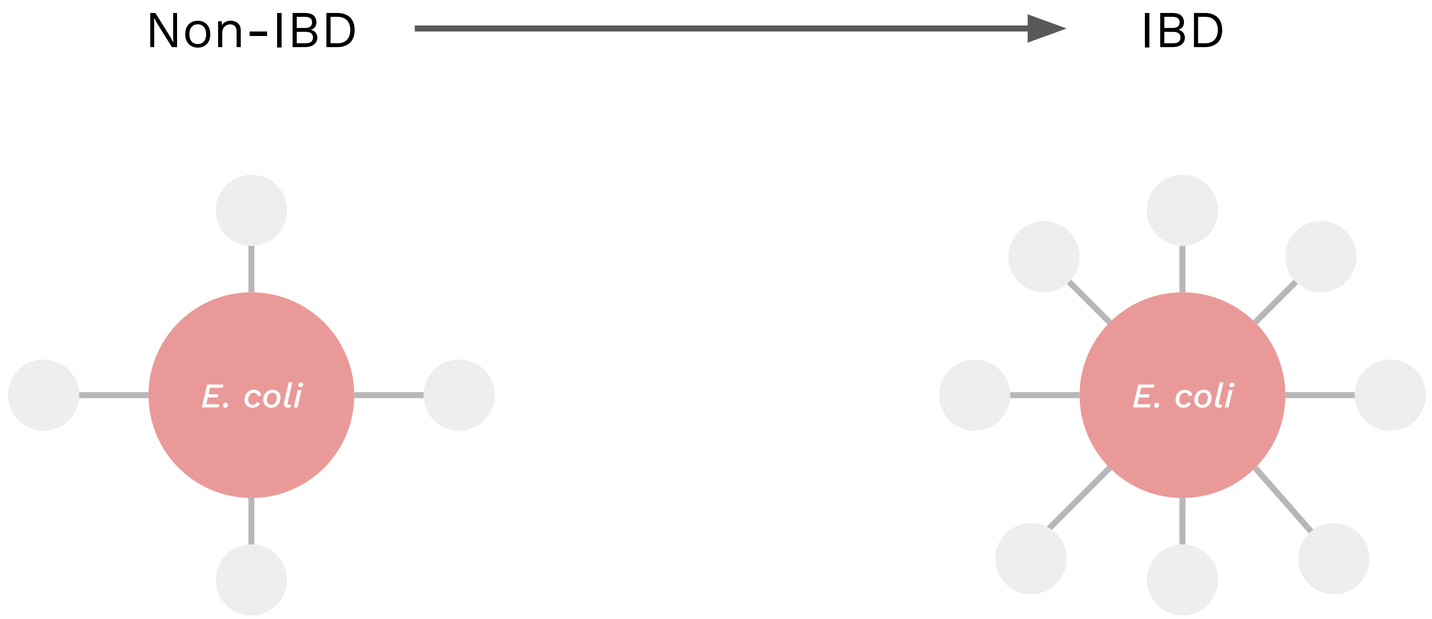
Pseudomonadota + Bacillota



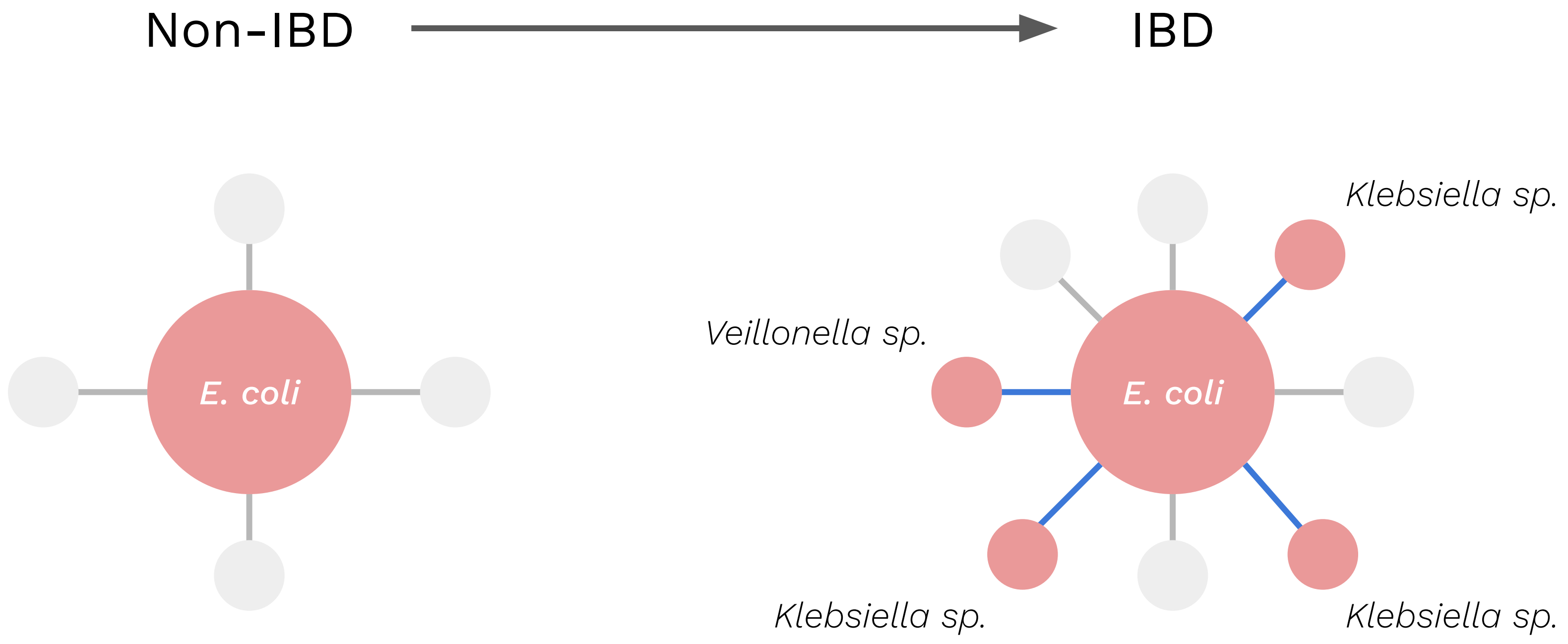
Bacteroidota



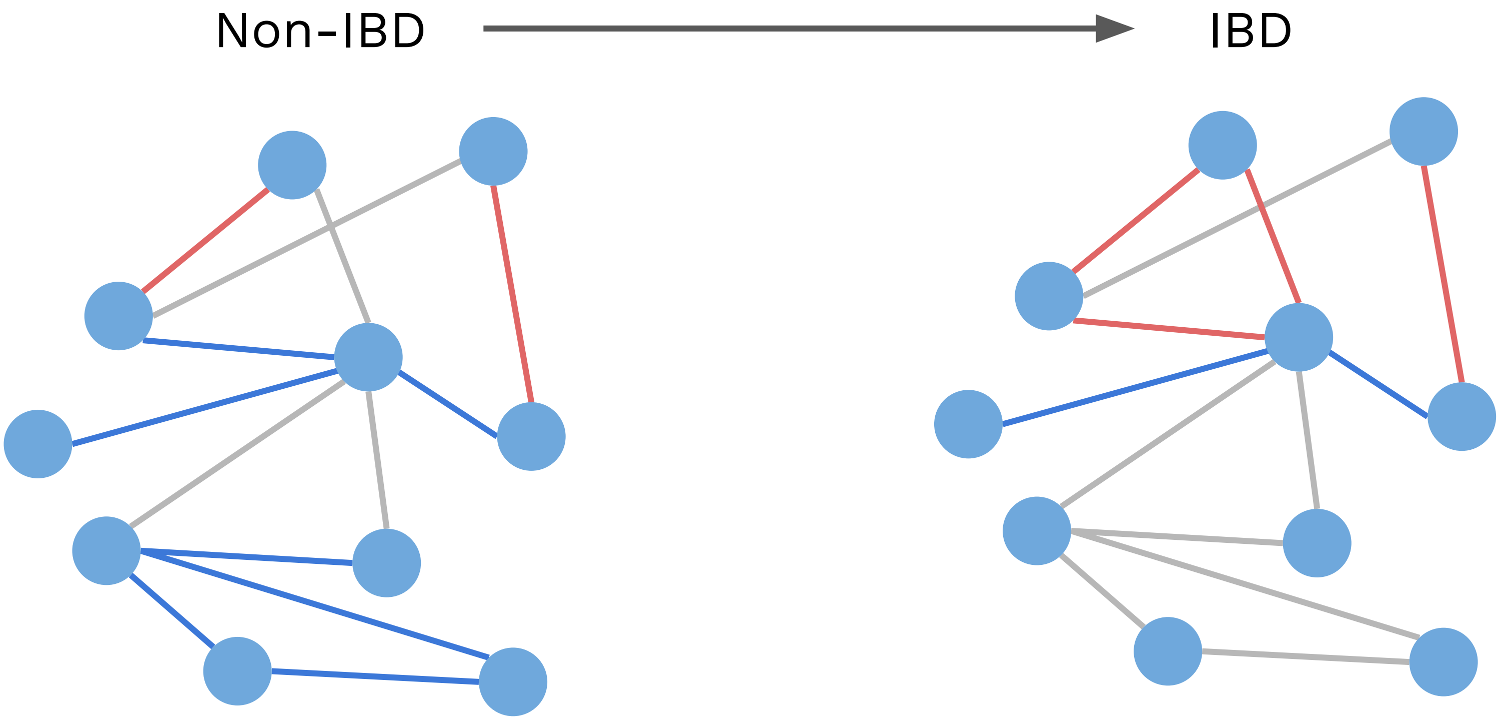
Networks → Model → Network-level features → Validation



E. coli emerges as a **central node** in individual IBD networks



E. coli becomes **positively associated** with pathobionts of the **oral microbiota**



Associations within the **Bacteroidota** module become **less positive**

Conclusion

- Although the graph-based models **were not the best at classifying patients**, we were able to **extract relevant network-level features** that offered a more **holistic view** of the differences in the gut microbiome of IBD and non-IBD patients.

Thank you!

MoNA website

multiomics-analytics-group.github.io

MoNA GitHub

github.com/multiomics-analytics-group

Acknowledgements

DTU Health Tech
Department of Health Technology

Multi-omics Network Analytics

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